DEFORESTATION

INTRODUCTION:

- Deforestation is the clearance or the removal of a forest or stand of trees from land which is then converted to a non-forest use.
- Deforestation can involve conversion of forest land to farms, ranches, or urban use. The most concentrated deforestation occurs in tropical rainforests.
- About 31% of Earth's land surface is covered by forests. Between 15 - 18 million hectares of forest, an area the size of Belgium are destroyed every year, on average 2,400 trees are cut down each minute.

CAUSES:

1) Mining : The increase of mining on tropical forests is furthering damage due to the rising demand and high mineral prices. These projects are often accompanied by large infrastructure construction, such as roads, railways, and power systems. This contributor to deforestation is putting additional pressure on our forests and freshwater ecosystems.

2) Paper : America, China, Japan, and Canada make up more than of the world's paper production —400 million tons a year. Approximately 640 million trees represent the paper that's thrown away each year, according to the Environment Paper Network.

3) Overpopulation :Due to overpopulation, more land is needed to establish housing and settlements. As well as many, many more roads and highways are being built in order to accommodate a larger sum of people driving. With more people that come with a large need for food and farmland to grow on and raise livestock—resulting in deforestation. Logging industries will cut down trees for furniture, paper, building materials, and many more products.

3) Logging : Wood-based industries such as paper, matchsticks, and furniture need a substantial quantity of wood. Lumber and charcoal are common examples of trees being used as fuel. Cooking and heating all around the world use these resources, and half of the illegal removal from forests is thought to be used as fuelwood. Large areas are also cleared to construct roads in order for large trucks to have entry to logging sites.

4) Agriculture Expansion : A major cause of deforestation is agriculture plantations. An increasing supply-demand for products such as palm oil and soybeans are driving producers to clear forests at an unnerving rate. Farmers often clear the land for cattle by using slash and burn techniques . Unfortunately, they will then use the property until the soil is completely degraded and repeat the process on a new patch of woodland (Jhuming culture).Cattle ranching and deforestation are strongest in Latin America. Over the past 40 years, forest area has reduced by almost an 40 percent. During the same period, 40 years, pasture regions and cattle population have grown significantly and rapidly.

5) Climate Change : Forests are essentially the lungs of our planet.Though. Forest loss is often caused by climate change. Tropical rainforests are extremely humid due to the water vapor released along with the oxygen. But when a forest is cut down, the humidity levels decrease and causes the remaining plants to dry out. For example, drying out our tropical rainforests increases fire damage. Fires can be both accidental and intentional but destroy forests quickly.

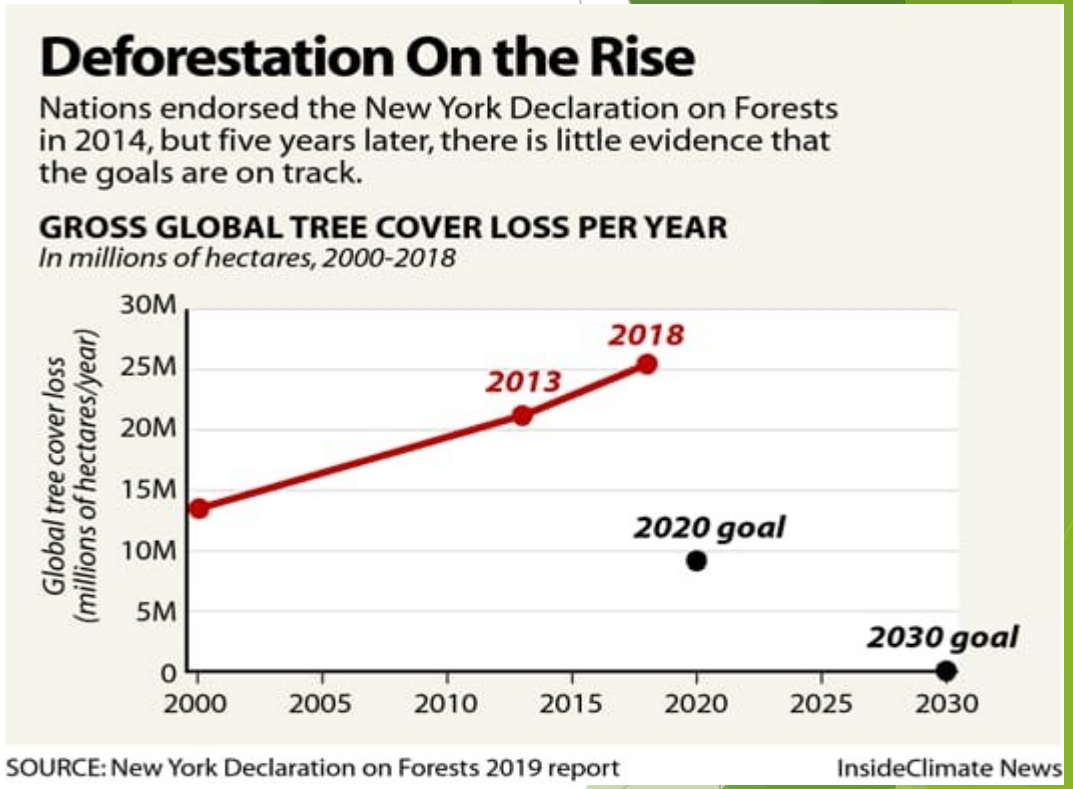
EFFECTS:

- Increased Greenhouse Gas Emissions: Trees help to mitigate carbon dioxide and other greenhouse gas emissions, but they become carbon sources once they're cut, burned, or otherwise removed. It's estimated that deforestation is responsible for around 20% of greenhouse gas emissions and 1.5 billion tons of carbon is released every year by tropical deforestation.
- Acidic Oceans:The oceans are becoming more acidic with an increased supply of carbon dioxide from deforestation and burning fossil fuels. Oceans are already 30 percent more acidic since the Industrial Revolution, putting ocean species and ecosystems at an extreme risk.
- Loss of Species: Orangutans, giant pandas, rhinos, and the Asian elephant are just a few of hundreds of endangered species due to deforestation. Removing trees thins the forest canopy which is meant to block sun rays during the day and holds in the heat at night. This damaging disruption leads to extreme temperature swings that are harmful to plants and animals. Many animals, insects, and plants lose their habitats and may become endangered and even go extinct.Though a few species are killed directly in forest clearances, many will face a slower death sentence due to a lack of food and breeding rates decline. White-cheeked spider monkeys are endangered largely specifically because of the enlargement of farmland and road construction.

- Because of water pollution from mining operations and agricultural runoff the giant otter is now endangered.It is estimated that we are losing 137 plant, animal, and insect species every day which equates to 50,000 species a year. It's also been estimated that 40% of the animal and plant species in Southeast Asia could be wiped out in the 21st century. A recent study of the Brazilian Amazon predicts that up to 90% of extinctions will occur in the next 40 years.
- Flooding and Erosion: Without trees to secure fertile soil, erosion often occurs and sweeps sacred land into nearby rivers.Trees are also crucial for our local water cycles as they assist in returning water vapor to the atmosphere. Forests serve as nature's water purification plants and as the rain water percolates within the soil and is held in place by intricate root structures of many layers of trees. When the protective forest canopy and roots have been destroyed, the soil will lose its proportions to retain water and is washed away into rivers and streams.
- Life Quality Decrease: Millions of people in the world depend on forests for hunting, small-scale agriculture, gathering, and medicine. Common materials we use every day such as latex, cork, fruit, nuts, natural oils, and resins are found in the tropical forests. Deforestation disrupts the lives of millions of people. In Southeast Asia, deforestation has contributed to social conflict and migration. Poor people from Brazil have been lured from their villages to soy plantations where they can be abused and forced, at gunpoint, to work under inhumane conditions

REPORTS :

- The World Bank estimates that about 3.9 million square miles (10 million square km) of forest have been lost since the beginning of the 20th century. In the past 25 years, forests shrank by 502,000 square miles (1.3 million square km) — an area bigger than the size of South Africa.
- In 2018, The Guardian reported that every second, a chunk of forest equivalent to the size of a soccer field is lost.
- Nature Scientific Reports if deforestation continue in current rate in the next 20 – 40 years, it can trigger a full or almost full extinction of humanity.
- According to FAO(food and agriculture Orgnization) Nigeria has the world's highest deforestation rate of primary forests. It has lost more than half of its primary forest in the last five years



Constitutional provisions for protection of forest :

The 42nd Amendment also introduced a new Directive Principle of State Policy [Article 48-A] and a Fundamental Duty [51 (A) (g)] for the protection and improvement of the forests. These provisions provide as under:

1.Article 48-A – Protection and improvement of environment and safeguarding of forests and wildlife. The State shall endeavor to protect and improve the environment and to safeguard the forests and wildlife of the country.

2. Article 51(A) (g) – It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wildlife, and to have compassion for living creatures.

Current Forest and Wildlife Legislations in India

The important forest legislations in India are:

1. The Indian Forest Act, 1927
2. The Wildlife Protection Act, 1972
3. The Forest Conservation Act, 1980
4. The Scheduled Tribes and other Traditional Forest Dwellers Act, 2006

ROLE OF NHRC IN PREVNTATION OF FOREST

There are various attempts of NHRC in prevention of deforestation ,following are the examples

- NHRC chairperson calls for combating 'ECO-TERRORISM'.
- Commission Attempts to Balance the Needs of Environmental Protection and the Rights of the Van Gujjars in uttar Pradesh.
- NHRC notices to the Centre and the Governments of Rajasthan and Haryana over the reported massive deforestation in Aravali Hills causing dust storm and health concerns in Delhi-NCR

Role of Non-Governmental Organisations (NGOs)

- Green Peace international is a NGO that mainly focuses on the environmental problems that are occurring in Southeast Asia.
- The World Wide Fund for Nature (WWF) is an international non-governmental organization founded in 1961 that works in the field of wilderness preservation and the reduction of human impact on the environment.It was formerly named the World Wildlife Fund .

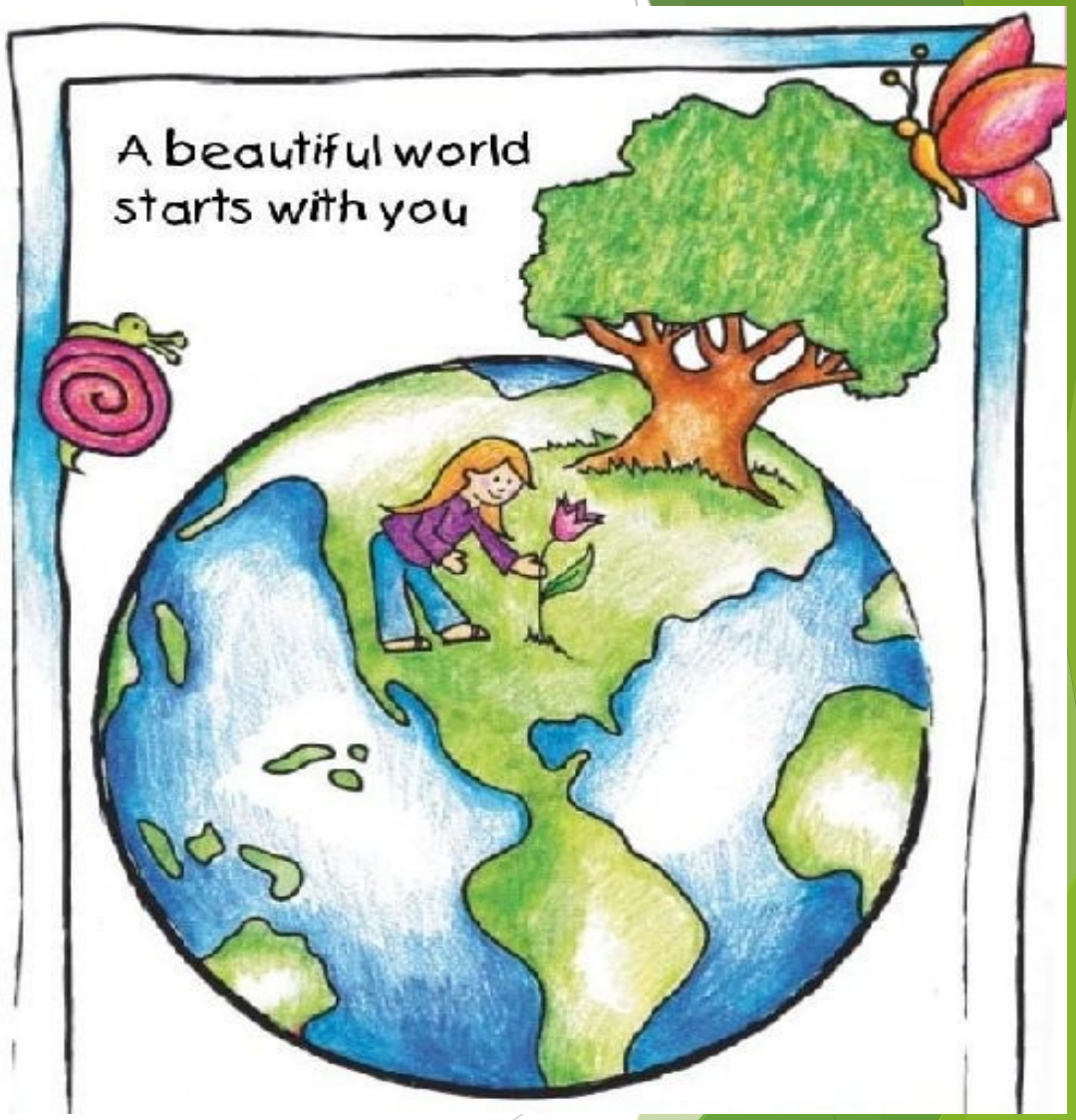
- The World Resources Institute (WRI) is a global research non-profit organization established in 1982 .WRI maintains international offices in the United States, China, India, Brazil, Indonesia, Mexico, and Brazil. organization's mission is to promote environmental sustainability, economic opportunity, and human health and well-being.

DEFORESTATION CONSERVATION GROUPS

- AMAZON WATCH
- CANADIAN FORESTRY ASSOCIATION
- CONSERVATION INTERNATIONAL
- RAINFORESTATION ACTION NETWORK
- RAINFOREST ALLIANCE
- TREES FOR THE FUTURE

CONCLUSION:

The various central legislations ,the international organizations and the NGOs for forest conservation and wildlife protection provide a powerful legal framework for protection of wildlife, establishment of protected areas, management of habitats, regulation and control of hunting and trade in parts and products derived from wildlife.
Although many organisations work for the conservation of nature,but it is responsibility of every individual to understand and take efforts in field of protection and educate others about the environment effects of deforestion.



THANK YOU

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